

QITAI, China

WMO: 513790

Lat: 43.9953N

Lon: 89.6044E

Elev: 795

StdP: 92.13

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.2	-27.4	-33.5	0.2	-30.0	-30.3	0.2	-27.2	6.3	-16.0	5.5	-18.3	2.6	160	0.531

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	34.4	17.8	32.9	17.4	31.4	16.9	19.5	30.1	18.7	29.2	18.0	28.4	2.2	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.3	12.7	22.2	15.3	12.0	21.6	14.4	11.3	21.6	59.2	29.7	56.4	29.3	54.1	28.3	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.3	6.0	5.1	DB	-32.3	37.9	3.1	2.0	-34.5	39.3	-36.4	40.5	-38.1	41.6	-40.4	43.0	(4)
(5)			WB	-32.4	21.1	3.1	1.6	-34.7	22.2	-36.5	23.1	-38.2	24.0	-40.5	25.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	-16.3	-12.7	-0.6	10.8	16.8	21.5	23.2	22.0	16.5	8.2	-1.7	-12.6	(6)	
	DBStd	14.74	5.10	4.97	7.39	5.02	3.99	3.01	2.68	3.08	4.07	4.58	5.86	5.22	(7)	
	HDD10.0	2982	816	635	333	51	3	0	0	0	4	89	350	701	(8)	
	HDD18.3	4803	1074	868	588	229	76	8	1	6	81	313	600	959	(9)	
	CDD10.0	1649	0	0	3	74	214	346	408	372	198	34	0	0	(10)	
	CDD18.3	430	0	0	0	2	28	104	151	119	24	1	0	0	(11)	
	CDH23.3	5517	0	0	0	74	463	1242	1771	1540	415	13	0	0	(12)	
	CDH26.7	2237	0	0	0	13	134	488	797	677	126	2	0	0	(13)	
(14) Wind	WSAvg	2.5	2.2	2.2	2.5	3.2	3.0	2.8	2.7	2.6	2.6	2.4	2.3	2.1	(14)	
(15) Precipitation	PrecAvg	175	4	5	9	19	18	25	25	16	21	15	10	7	(15)	
	PrecMax	326	16	18	23	48	49	60	79	51	57	35	32	17	(16)	
	PrecMin	90	0	0	0	4	1	0	1	2	1	1	2	1	(17)	
	PrecStd	47	4	4	6	11	12	16	21	12	14	8	6	5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.3	1.8	20.1	28.3	32.1	35.1	37.3	36.3	32.9	24.9	15.9	1.6	(19)	
		MCWB	-3.3	-0.9	8.9	12.5	15.3	17.6	19.3	17.7	15.6	11.2	7.2	-1.1	(20)	
	2%	DB	-4.1	-0.8	16.5	25.3	29.9	33.0	34.7	34.1	30.0	21.9	11.7	-1.1	(21)	
		MCWB	-5.4	-2.9	8.2	10.9	14.5	17.4	18.6	17.4	14.8	10.3	5.2	-2.5	(22)	
	5%	DB	-6.4	-2.7	13.5	23.0	28.2	31.4	32.9	32.4	27.8	19.4	8.9	-3.1	(23)	
		MCWB	-7.4	-4.3	7.0	10.2	14.0	16.7	18.0	17.1	13.9	9.5	3.9	-4.3	(24)	
	10%	DB	-8.6	-4.5	10.4	20.6	26.0	29.7	31.2	30.7	25.6	17.0	6.2	-4.9	(25)	
		MCWB	-9.4	-5.9	5.5	9.3	13.0	16.4	17.6	16.6	13.1	8.6	2.6	-5.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.8	-0.4	10.1	13.4	17.1	19.8	21.4	19.9	17.1	12.1	7.6	-0.6	(27)	
		MCDB	-1.7	1.3	17.9	25.9	28.9	31.1	33.0	29.5	28.1	23.3	14.9	0.9	(28)	
	2%	WB	-5.3	-2.6	8.5	12.0	15.6	18.7	20.0	18.8	15.7	10.9	5.5	-2.4	(29)	
		MCDB	-4.2	-1.1	15.6	22.0	27.2	29.3	30.9	30.0	27.5	20.0	10.9	-1.1	(30)	
	5%	WB	-7.4	-4.2	7.2	10.9	14.6	17.9	19.1	18.0	14.7	9.8	4.2	-4.2	(31)	
		MCDB	-6.4	-2.7	13.1	20.5	25.5	28.2	29.4	29.4	26.0	18.1	8.5	-3.2	(32)	
	10%	WB	-9.4	-5.8	5.7	10.0	13.7	17.2	18.4	17.3	13.7	8.8	2.8	-5.8	(33)	
		MCDB	-8.6	-4.6	10.4	19.3	24.0	27.1	28.4	28.3	24.1	16.5	6.1	-4.8	(34)	
(35) Mean Daily Temperature Range	MDBR		12.1	12.9	11.7	13.3	13.0	12.8	13.1	14.2	14.9	13.3	10.4	10.7	(35)	
	5% DB	MCDBR	12.5	13.4	15.1	17.6	16.7	16.7	17.1	17.5	18.5	16.9	14.0	10.8	(36)	
		MCWBR	11.1	11.4	7.9	7.1	6.0	6.0	5.7	6.0	7.2	7.8	8.5	9.3	(37)	
	5% WB	MCDBR	12.1	12.7	13.9	14.3	14.2	13.1	14.0	15.0	15.9	15.4	13.2	10.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.332	0.367	0.412	0.459	0.454	0.471	0.452	0.428	0.401	0.385	0.353	0.327	(40)	
	taud		2.120	2.074	1.992	1.924	1.986	2.012	2.105	2.153	2.150	2.109	2.128	2.138	(41)	
	Ebn at Noon		755	796	808	803	821	807	818	824	814	764	720	717	(42)	
	Edh at Noon		99	125	155	180	175	171	154	142	131	119	98	89	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.97	2.96	4.33	5.68	6.65	6.83	6.54	6.04	5.15	3.62	2.25	1.63	(44)	
	RadStd		0.10	0.13	0.24	0.35	0.33	0.38	0.30	0.30	0.22	0.18	0.18	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		+0.50	N/A	N/A	N/A	N/A	-0.60	N/A	N/A	+98	+49

Nomenclature: See separate page